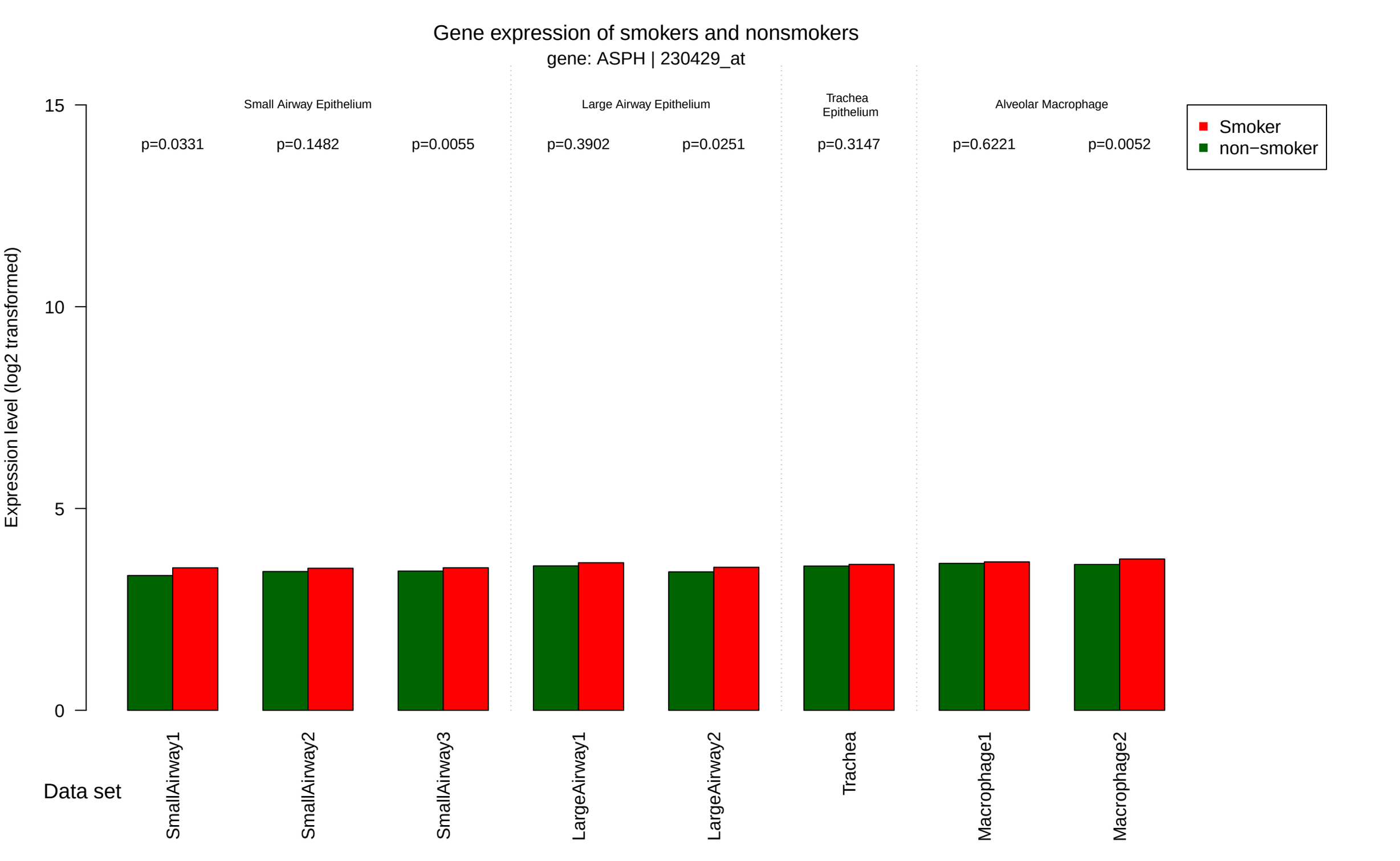
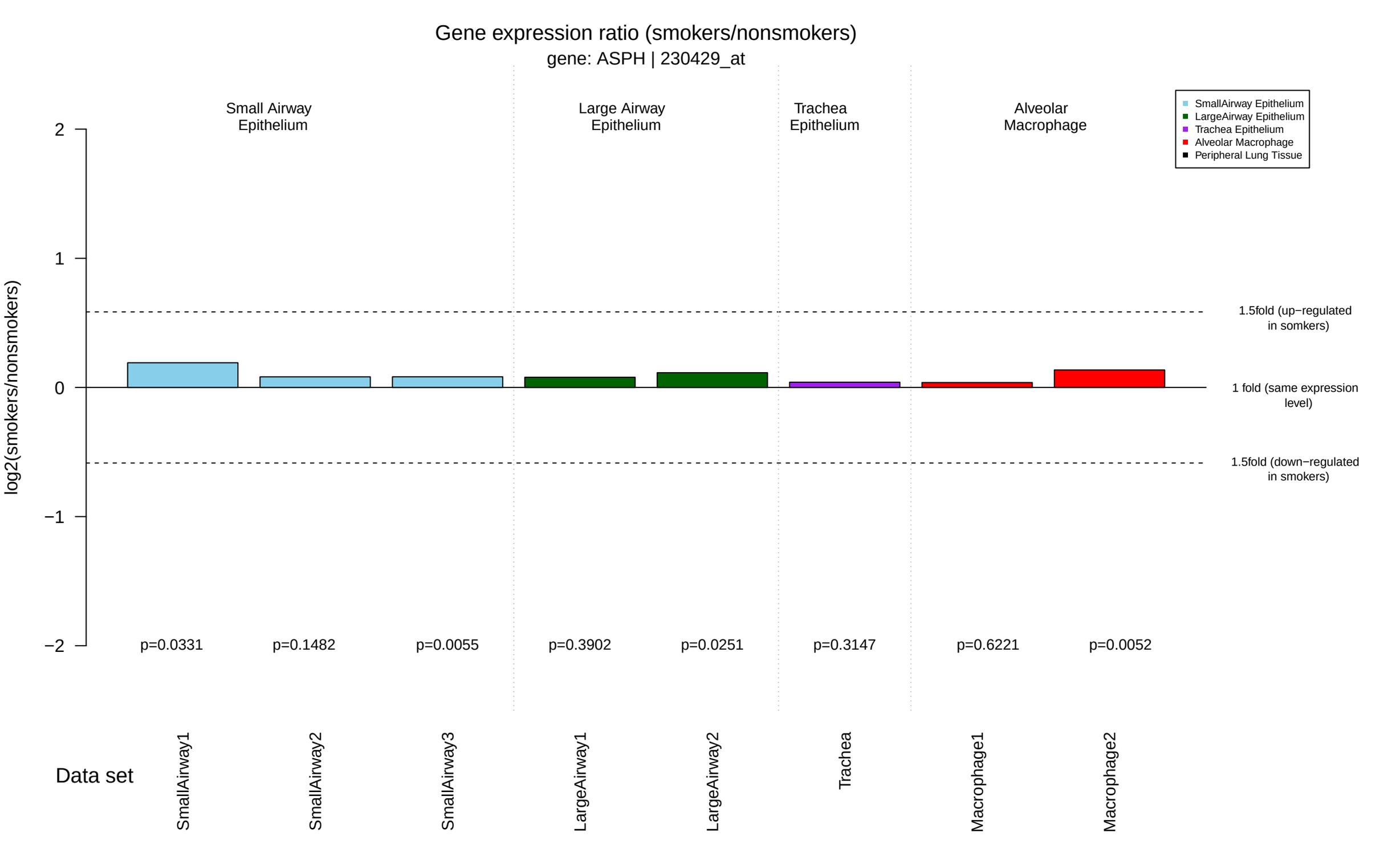
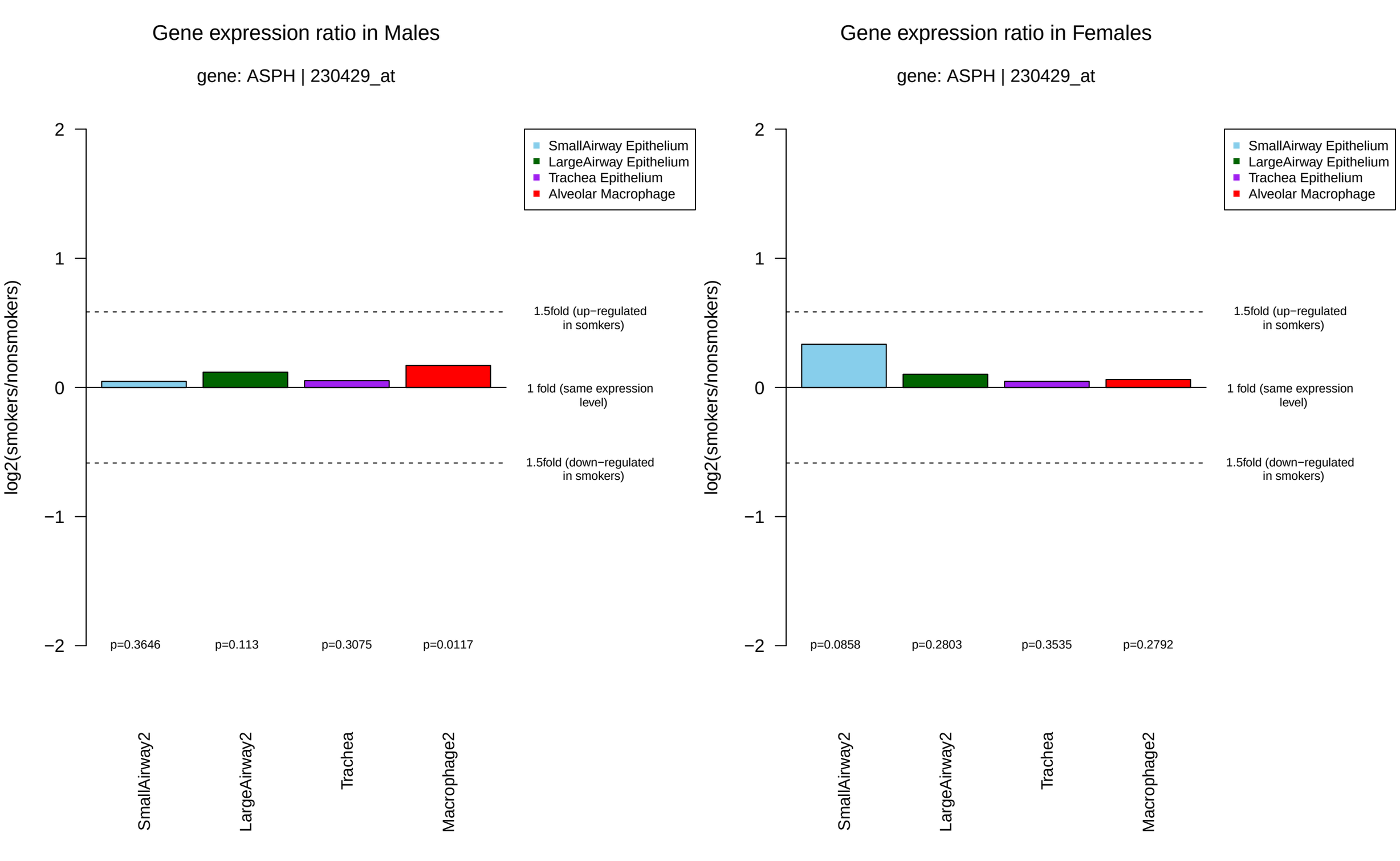
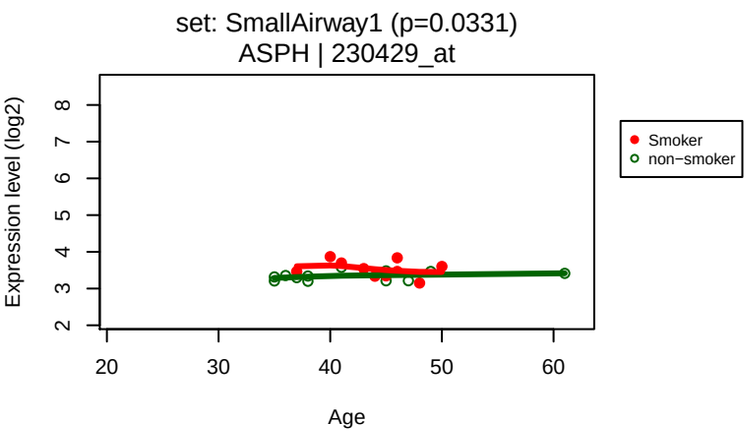
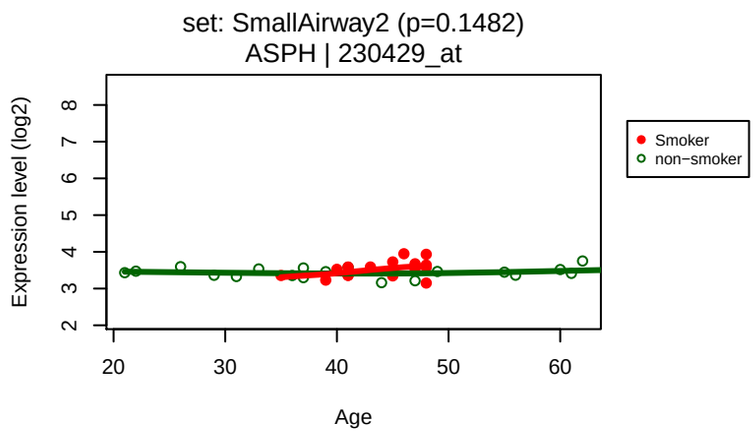
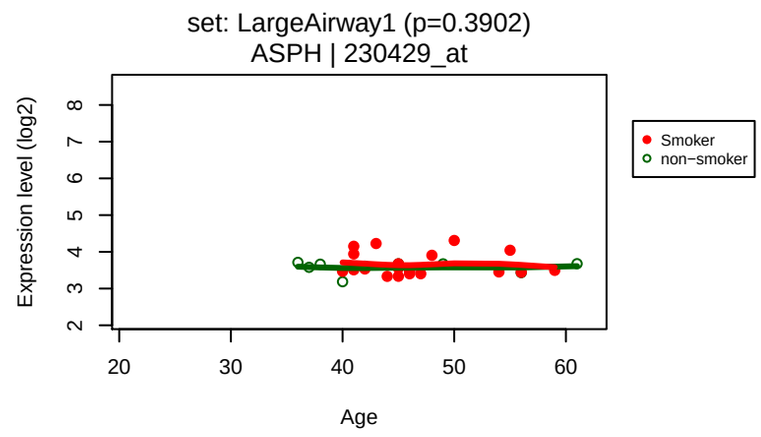

Gene expression of smokers and nonsmokers
gene: ASPH | 230429_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0331
p=0.1482
p=0.0055
p=0.3902
p=0.0251
p=0.3147
p=0.6221
p=0.0052
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

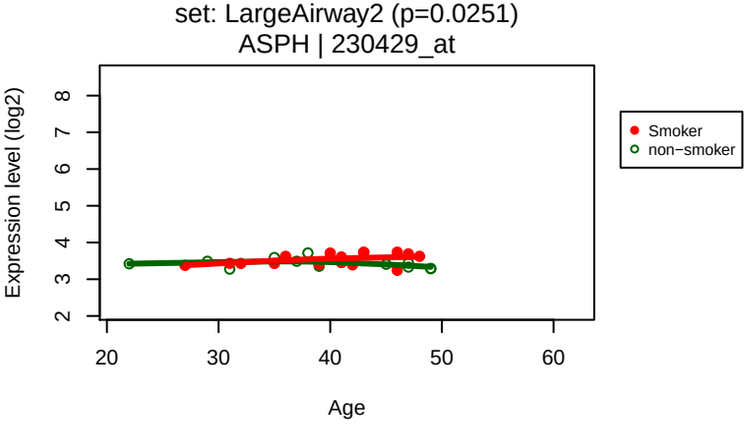

Gene expression ratio (smokers/nonsmokers)
gene: ASPH | 230429_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0331
p=0.1482
p=0.0055
p=0.3902
p=0.0251
p=0.3147
p=0.6221
p=0.0052
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

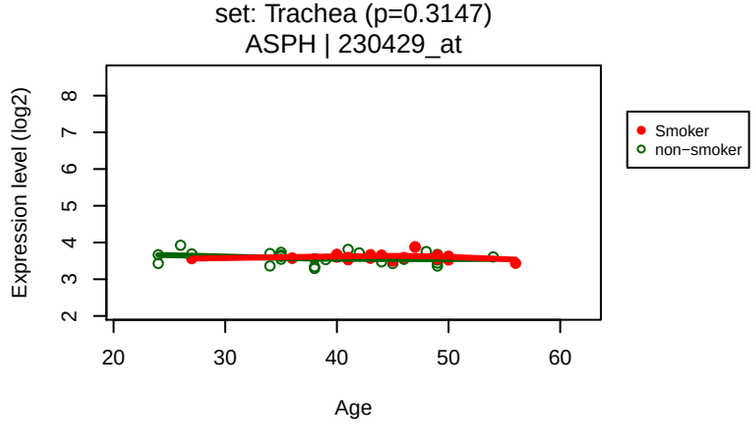

Gene expression ratio in Males
gene: ASPH | 230429_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3646
p=0.113
p=0.3075
p=0.0117
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASPH | 230429_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0858
p=0.2803
p=0.3535
p=0.2792
SmallAirway2
LargeAirway2
Trachea
Macrophage2

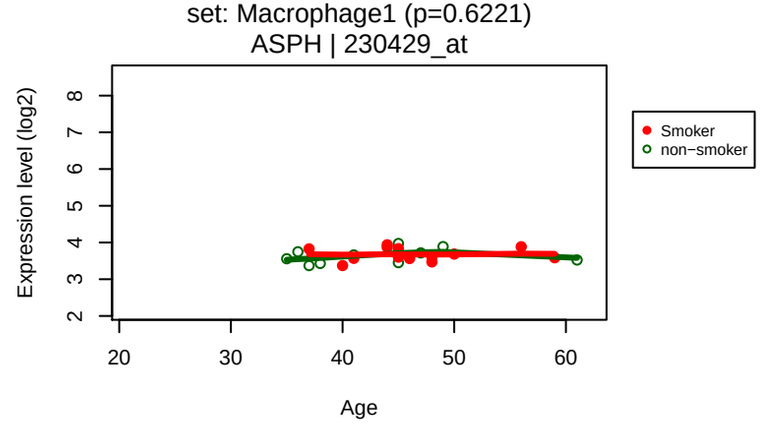

set: SmallAirway1 (p=0.0331)
ASPH | 230429_at
Expression level (log2)
Age
Smoker
non−smoker

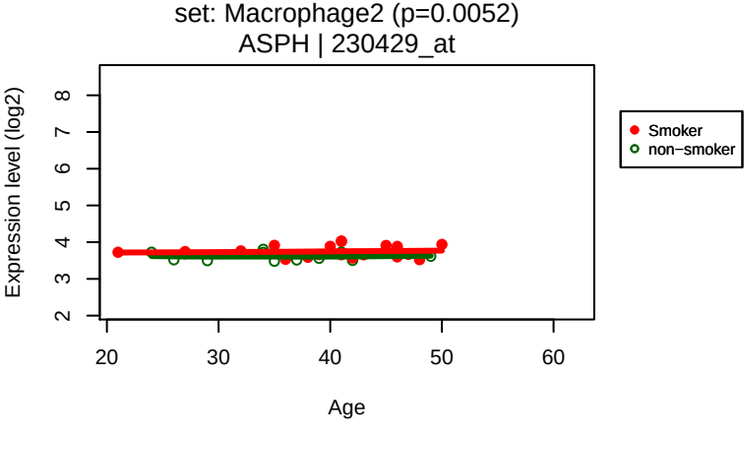

set: SmallAirway2 (p=0.1482)
ASPH | 230429_at
Expression level (log2)
Age
Smoker
non−smoker

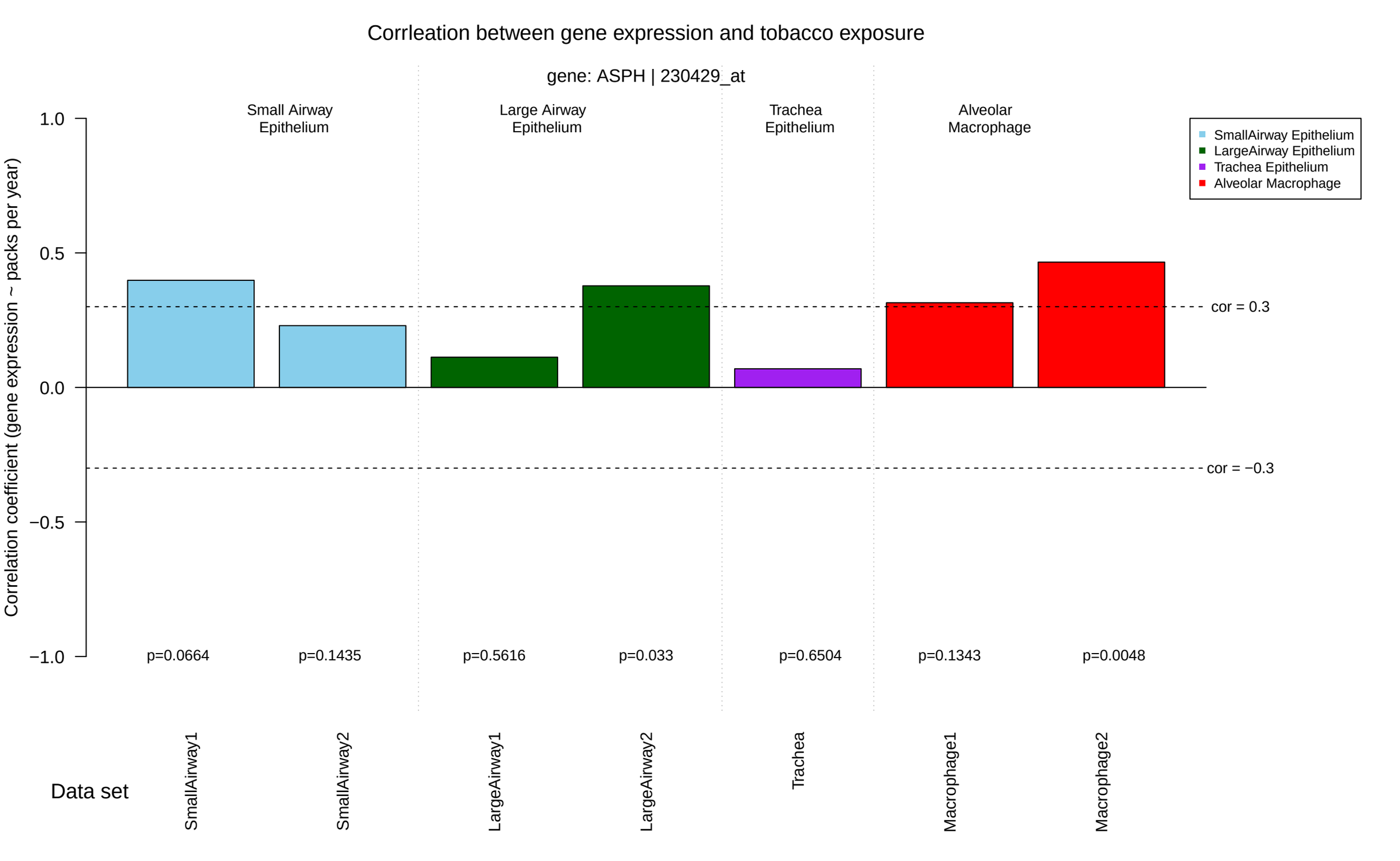

set: LargeAirway1 (p=0.3902)
ASPH | 230429_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0251)
ASPH | 230429_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3147)
ASPH | 230429_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6221)
ASPH | 230429_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0052)
ASPH | 230429_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASPH | 230429_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0664
p=0.1435
p=0.5616
p=0.033
p=0.6504
p=0.1343
p=0.0048
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2